

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

IO Organization

Lecture No.- 1

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Recap of Previous Lecture



Topic

Floating-Point Numbers

↳ doubts ?

Topic

Biased Exponent

Topic

Number Range

Topic

IEEE-754 Floating Point Representation

Topic

Denormalized Number

Topics to be Covered



Topic

Peripheral Device

Topic

I/O vs Memory Buses

Topic

Memory Mapped I/O vs I/O Mapped I/O

Topic

Asynchronous Data Transfer

#Q. The value of a float type variable is represented using the single-precision 32-bit floating point format IEEE-754 standard that uses 1bit for sign, 8 bits for biased exponent and 23 bits for mantissa. A float type variable X is assigned the decimal value of -14.25 . The representation of X in hexadecimal notation is

A ✓ C1640000H

B 416C0000H

C 41640000H

D C16C0000H

[NAT]



#Q. Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127.

S: **1** E: 10000001 F: 111100000000000000000000

Here S, E and F denote the sign, exponent and fraction components of the floating-point representation.

The decimal value corresponding to the above representation (rounded to 2 decimal places) is **- 7.75**



[NAT]



#Q. The format of the single-precision floating-point representation of a real number as per the IEEE 754 standard is as follows:

Sign	Exponent	mantissa
------	----------	----------

Which one of the following choices is correct with respect to the smallest normalized positive number represented using the standard?

- A. exponent = 00000001 and mantissa = 00000000000000000000000000000001
- B. ✓ exponent = 00000001 and mantissa = 00000000000000000000000000000000
- C. exponent = 00000000 and mantissa = 00000000000000000000000000000000
- D. exponent = 00000000 and mantissa = 00000000000000000000000000000001



Topic : Booth Multiplication

Example:

Multiplicand: 10111

Multiplier: 100110

1. Add a zero in right side of multiplier
2. for each consecutive 2 bits in multiplier

$\begin{array}{l} \rightarrow 00 \} \text{do nothing} \\ \rightarrow 11 \} \\ \rightarrow 01 \Rightarrow \text{Addition} \\ \rightarrow 10 \Rightarrow \text{subtraction} \end{array}$

$\rightarrow$ reduces no. of additions (operations) required for multiplication
 $\rightarrow$ works on 2^s complement numbers

10111
100110

sub.
add.
sub

3 operations

multiplicand:- 101101100
multiplier:- 100110010

5 operations

[NAT]

[2025 : 1M]



#Q. The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm:

M: 1100 1101 1110 1101

and Q: 1010 0100 1010 1010 ○

The total number of addition and subtraction operations to be performed is 13. (Answer in integer)



I/O organization



Topic : Peripheral Device

/ I/O device



all the devices which are externally connected to CPU
except main memory.

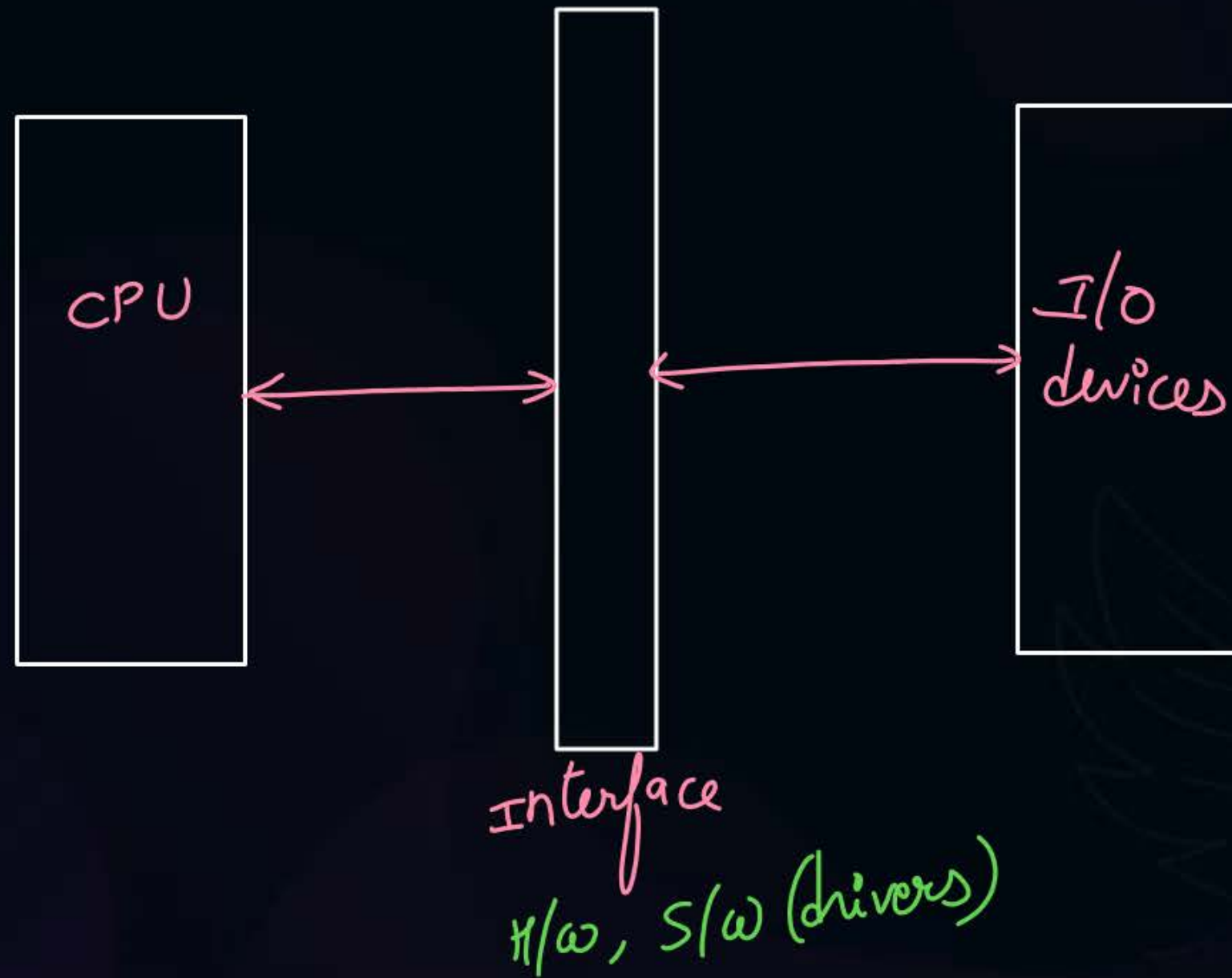
I/O devices

├→ Input devices
├→ output "
└→ storage "





Topic : CPU Connected to IO Directly?





Topic : Need For Interface

1. Peripherals are electromechanical or electromagnetic devices; and their manner of operation is different from the operation of the CPU and memory. Which are electronic devices. So conversion of signal required.
2. The data transfer rate of peripherals is usually slow. So synchronization is required.
3. Data codes and format in peripherals differ from the word format in the CPU and memory. So conversion of formats is required.
4. The operating modes of peripherals are different from each other and each must be controlled so a peripheral does not disturb the operation of other peripherals.



Topic : IO vs Memory Buses

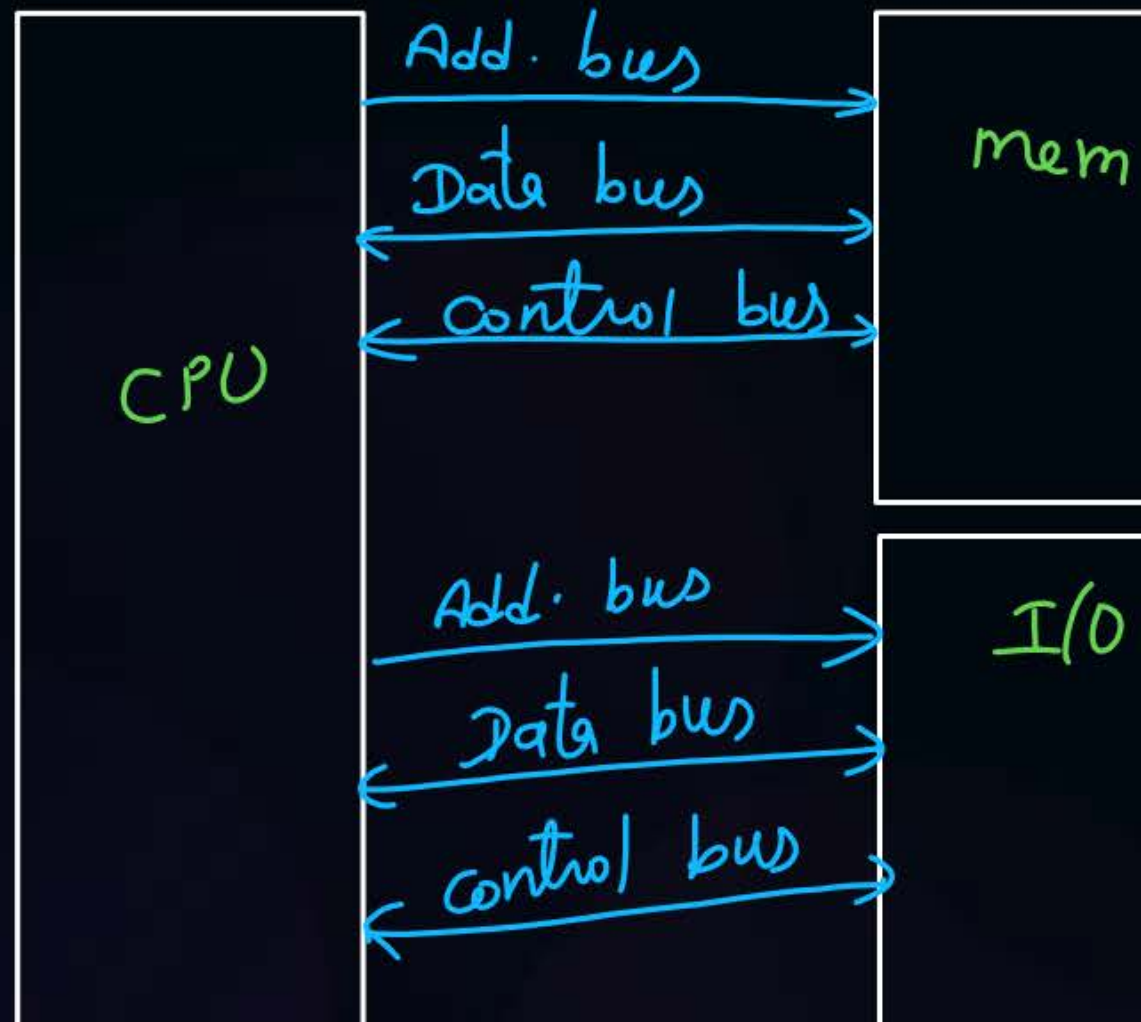


How CPU should connect to mem. and I/O through system buses





Topic : 1. Separate Buses for Both

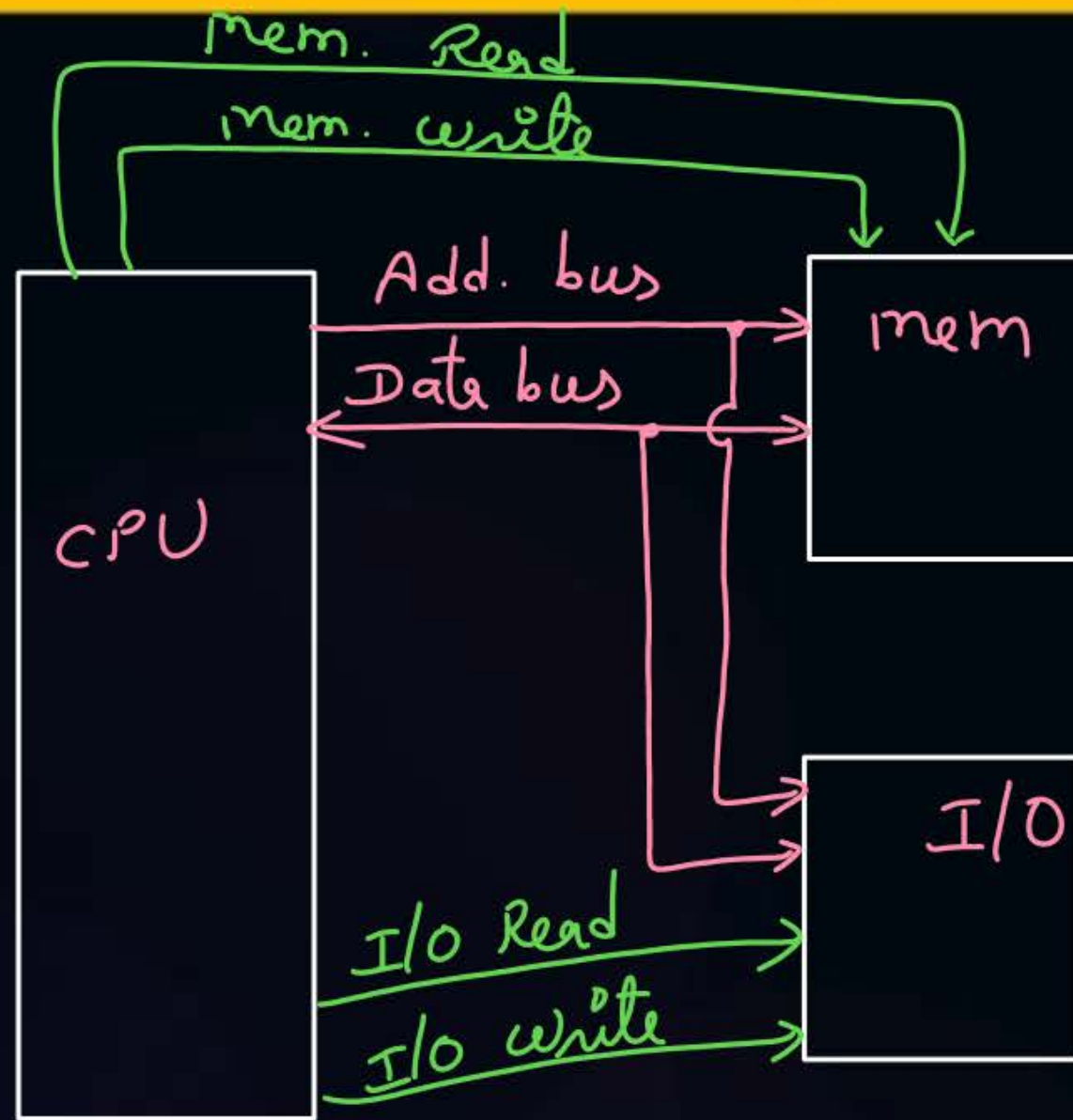


→ expensive

→ not practically used



Topic : 2. Common Data, Address Bus

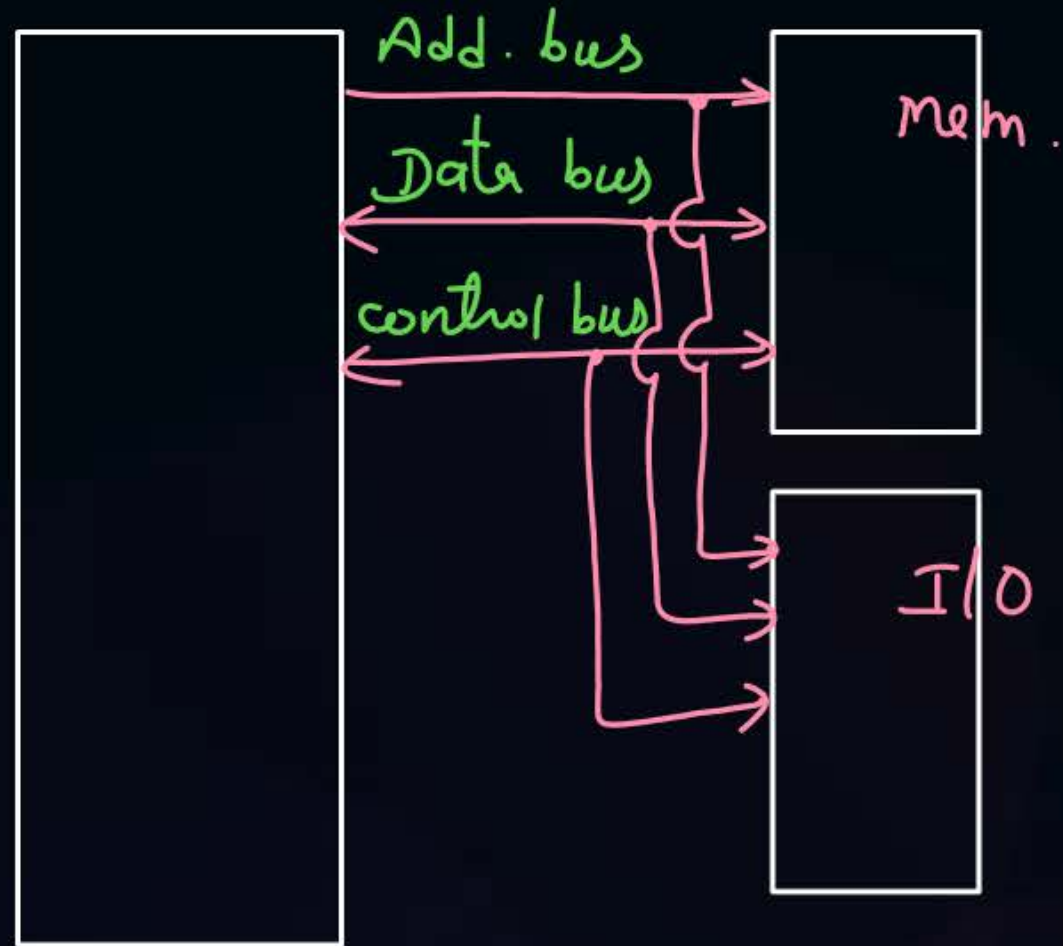


→ Practically implemented

Isolated I/O
or
I/O mapped I/O
or
Port mapped I/O



Topic : 3. Common Address, Data & Control Bus



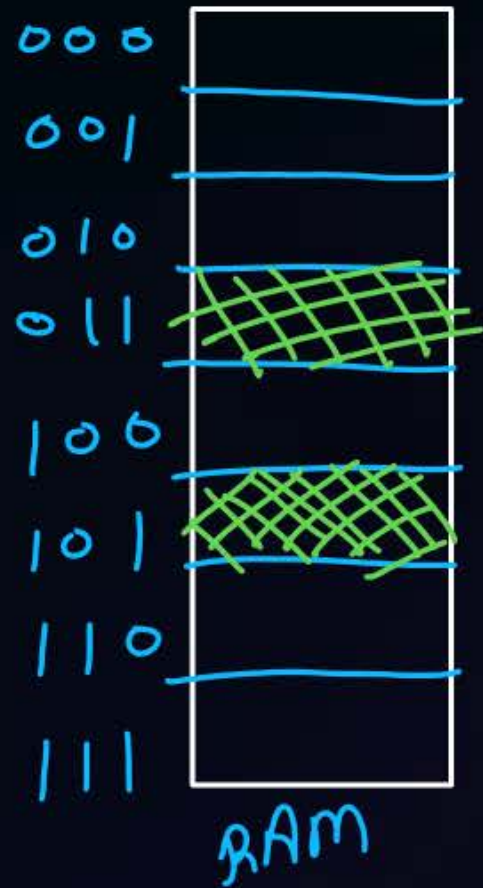
memory mapped I/O



Topic : 3. Common Address, Data & Control Bus

ex:- 8 bytes byte addressable RAM
2 I/O devices $\Rightarrow$ d1 and d2

2 addresses from mem. are assigned to I/O devices



	assigned address
d ₁	101
d ₂	011



Topic : Memory Mapped IO vs IO Mapped IO

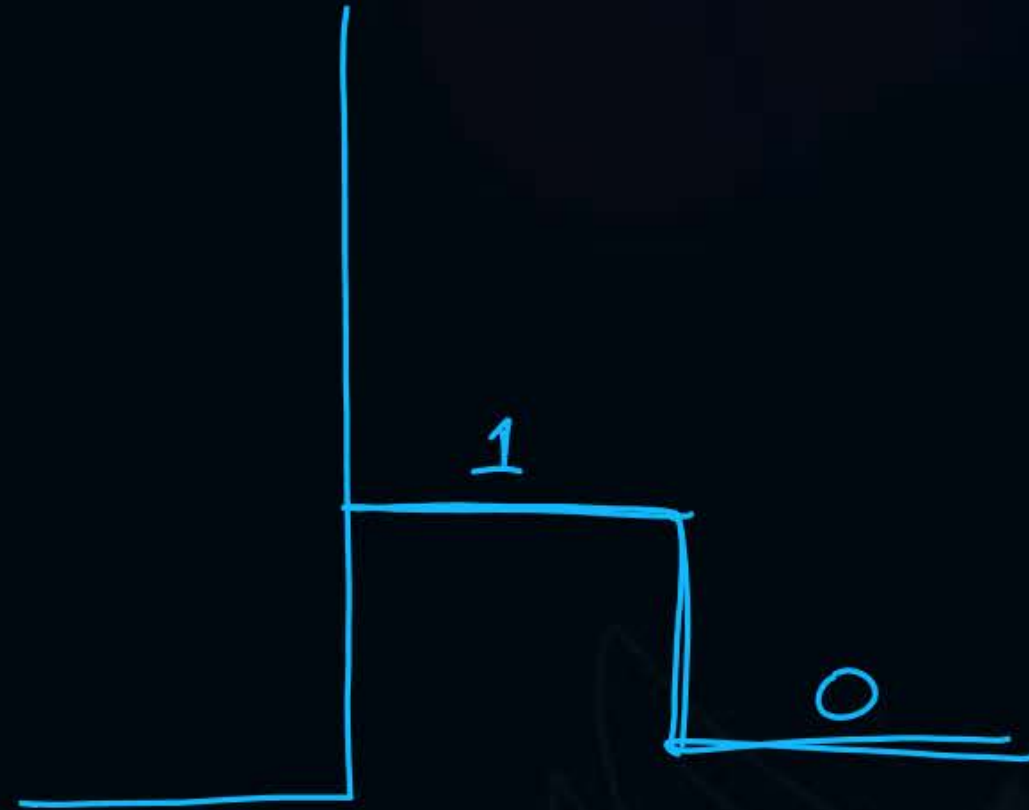
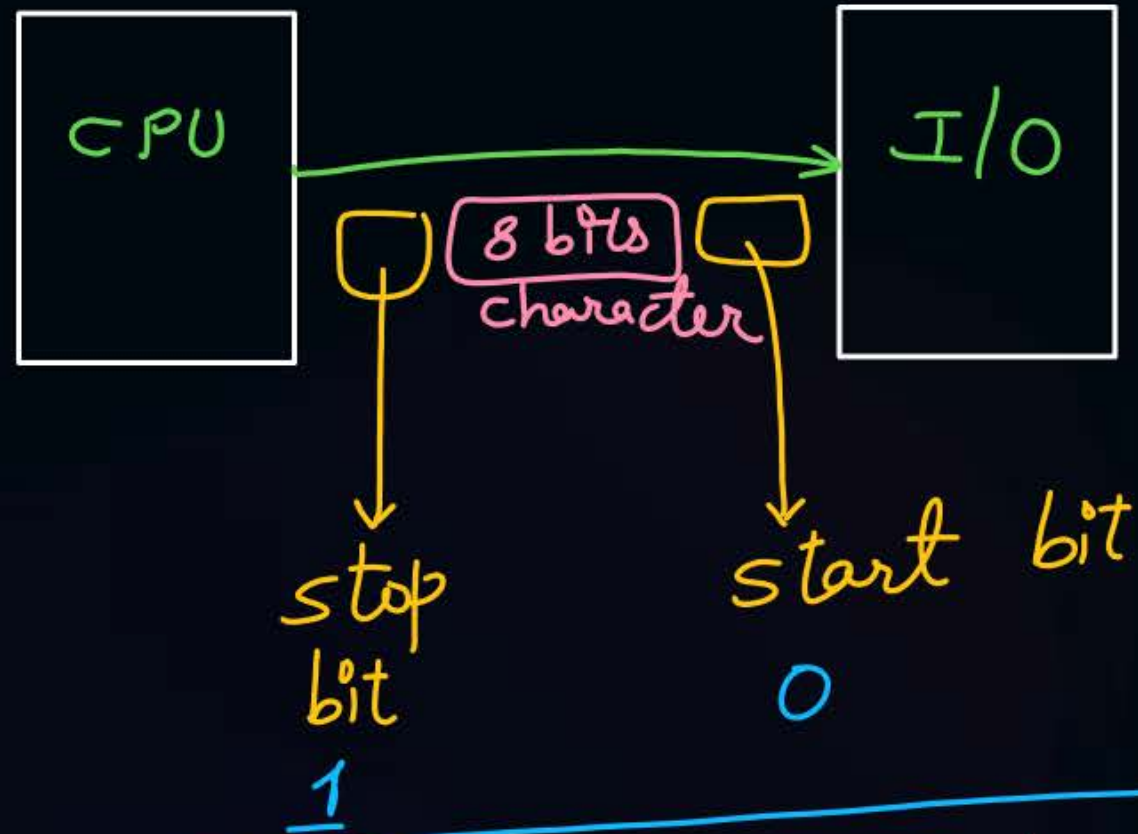
Memory Mapped IO	IO Mapped IO
1. Memory wastage	1. No Memory wastage
2. All Memory access instructions used for IO access also	2. IO access instructions and memory access instructions are different
3. No separate address space for IO	3. IO have their own separate address space
4. More Instructions for IO access	4. Less Instructions for IO access
5. More addressing modes for IO access	5. Less addressing modes for IO access
6. More devices can be connected to system	6. Less devices can be connected to system
7. ALU Operation can be performed on IO data directly	7. ALU Operation can not be performed on IO data directly



Topic : Asynchronous Data Transfer

(serial)

↳ one line b/w devices



To send one char of 8 bits, Total bits transferred = 10 bits

efficiency of transmission line = $\frac{8}{10} = 0.8$ or 80%

effective transfer rate of line = efficiency * Actual transfer rate

$$\text{Ans} = 800$$

#Q. How many 8-bit characters can be transmitted per second over 9600 bits/sec. serial communication link using a parity synchronous mode of transmission with 1 start bit, 8 data bits, 2 stop bits and 1 parity bit?

$$\text{for 1 char no. of bits sent} = 1 + 8 + 2 + 1 = 12 \text{ bits}$$

$$\frac{9600}{12} = 800 \text{ char/sec}$$

#Q. An asynchronous serial communication is employing 8 character bits, 1 parity bit, 2 start bits and 1 stop bit. To maintain a rate of 700 char/sec. The minimum transfer rate should be required is 8400 bits/sec?

for 1 char, no of bits sent = $1 + 2 + 1 + 8 = 12$ bits

for 700 char's " — " — = $12 * 700$
= 8400 bits

#Q. 8-bit characters can be transmitted using a parity synchronous mode of transmission with 1 start bit, 8 data bits, 2 stop bits and 1 parity bit.

1. What is the efficiency of the transmission line?
2. If the transfer rate of the line is 3000 bits per second, then effective transfer rate is?

no. of bits for each char = $1 + 8 + 2 + 1 = 12$ bits

1. efficiency = $\frac{8}{12} = \frac{2}{3} = 0.66 = 66.67\%$

2. $= \frac{2}{3} * 3000 \text{ bits/sec} = 2000 \text{ bits/sec}$



2 mins Summary



Topic

Peripheral Device

Topic

IO vs Memory Buses

Topic

Memory Mapped IO vs IO Mapped IO

Topic

Asynchronous Data Transfer



Happy Learning

THANK - YOU

